Kishan Narsingh Chhari

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Github · Portfolio

Profile

Embedded Systems & Robotics Engineer with hands-on experience in autonomous robots, UAVs, and embedded hardware end-to-end. M.Sc. candidate at ESIGELEC, France. Core strengths in C/C++, ROS2, ARM Cortex-M, RTOS, PCB design, IoT, and sensor fusion. Holds a granted patent in prosthetics. *Seeking: Embedded Systems / Robotics Engineer — Internship in France*

Work Experience

Assistant Technical Manager — Oct. 2023 — Aug. 2024
Learning Links Foundation, Mumbai

- Led **4–6** engineers to deliver a **sub-250g gesture-controlled drone** and **IoT humanoid robot** kit.
- Drove **250% growth** in organizational engagement and revenue.

Robotics Trainer — Jun. 2022 — Sep. 2023
Witty International School, Mumbai

- Built STEAM & Robotics curriculum from scratch — **300+ students**, **13+ projects**, national competition wins.

Projects

Self-Balancing Autonomous Robot — Nov. 2025 — Present
Arduino Nano, RPi 4, LiDAR, MediaPipe, ROS2, Custom PCB

- **Problem:** Multi-wheeled robots demand large footprints, restricting navigation in constrained indoor spaces.
- **Approach:** Two-wheeled inverted pendulum with LQR state-feedback, real-time disturbance rejection, 360° LiDAR avoidance via ROS2, and MediaPipe human following.
- **Implementation:** Custom 2-layer PCB, 3D-printed chassis, IMU–encoder fusion, 3-wire GPIO bus — sub-180 ms latency.

AI Voice Assistant & Surveillance Node — Sept. 2025
RPi 4, Python, OpenAI API, OpenCV

- **Problem:** Surveillance systems lack contextual intelligence, demanding constant human oversight.
- **Implementation:** Edge-AI pipeline on CPU-only RPi 4 — motion trigger → OpenCV inference → GPT narration, sub-2 s response, zero cloud dependency.

Autonomous Fixed-Wing UAV — 2022
Pixhawk, ArduPilot, Mission Planner, XFLR5

- **Problem:** Off-the-shelf UAVs sacrifice endurance for payload, limiting deployment range.
- **Implementation:** Power-optimised composite airframe with XFLR5 airfoil — **30 min endurance**, **500 g payload**, **50 m/s** cruise, autonomous waypoint nav via Mission Planner.

Education

M.Sc. Electronic Embedded Systems — 2024 — 2026
ESIGELEC, Rouen, France
Embedded Linux, RTOS, Robotics, IoT Architecture

B.Tech. Mechanical Engineering — 2018 — 2022
Lovely Professional University, Jalandhar
Specialization: Robotics & Automation

Skills

Languages: C/C++, Embedded C, Python, Embeded Java
Embedded: STM32, RPi, ESP32, FreeRTOS, Zephyr, Yocto, AUTOSAR, Device Drivers
Protocols: SPI, I2C, UART, CAN, DDS, MQTT, BLE
Robotics: ROS2, Nav2, Simulink, BLDC Control, ISO 26262, HIL Testing
Vision/AI: OpenCV, MediaPipe, Edge LLM Deployment
Tools: Git, CI/CD, Docker, GDB/JTAG, Keil, KiCad, SolidWorks (CSWA)
Hardware: Custom PCB, FPGA, 3D Printing, Oscilloscope

Publications & Patent

◇ **Patent: Bionic Arm** — IP India | 2023 | No. 202311042415
Servo-actuated 3D-printed prosthetic with EMG-based grip control at 50 Euros BOM cost.

◇ **Aerodynamic & Bending Analysis of Low-Speed Airfoils**
Materials Today: Proceedings — 2022

◇ **Design & Fabrication of Low-Cost Prosthetic Arm**
AIP Conference Proceedings — 2022

Awards

◇ **1st Place** — Robotics & UAV, IIT Bombay, IIT Ropar
◇ **3rd Place** — Aero-Vaccine Challenge, TECHFEST IIT Bombay
◇ **1st Place** — SolidWorks Design Challenge 2021

Languages English (Native) ●●●●●
French (B1 — In Progress) ●●○○○
Hindi (Native) ●●●●●